Department of Philology, Literature, and Linguistics

Master's Degree Program in Digital Humanities

GRADUATION THESIS

MAI(A) the Sound Be With You

MAIA-AV - A Ground-Up Audio-Visual Benchmark for Multimodal AI Assessment

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Academic Year 2026/2027

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Introduction

The rapid evolution of Vision-Language Models raises the need to determine whether their strong performance reflects genuine multimodal understanding or the exploitation of superficial correlations and linguistic biases. In this contex *MAIA: Multimodal AI Assessment* by Testa et al.(2025)[16, 15] was introduced as a native Italian benchmark for evaluating video-based reasoning in Vision-Language Models. Testa et al. investigate the extent to which VLMs coherently integrate visual and linguistic information. More specifically, it assesses whether model predictions are grounded in visual evidence rather than primarily driven by distributional regularities learned by the language component. This distinction is crucial because a model may answer a question correctly while failing to recognise the same information consistently when it is presented in a discriminative format. In such cases, the prediction may result from linguistic plausibility rather than genuine visual understanding. To address this issue, MAIA employs two aligned tasks: Visual Statement Verification and Open-Ended Visual Question Answering. Through these tasks, Testa et al. evaluate not only models accuracy, but also the robustness and consistency of models behaviour across different evaluation formats. The benchmark further defines twelve fine-grained reasoning categories to identify which competencies are used by the models and in which areas linguistic shortcuts, unimodal collapse, or hallucinations emerge. A limitation of the original framework is that the dataset was designed and evaluated primarily on the basis of visual information. Real-world videos, however, are inherently audiovisual: relevant meaning may also be conveyed through dialogue, environmental sounds, music, and other acoustic events. Recent multimodal and omnimodal architectures can process visual, auditory, and textual inputs, but the availability of multiple modalities does not guarantee that they will be integrated effectively. Models may over-rely on a single channel or be negatively affected by redundant or weakly relevant information. Starting from this problem, the study Deodati,Kaif et al.(2026): ”*Lost in Transcription: Investigating the Role of Audio Information for Video-based Visual Statement Verification*”[4] extends the original MAIA research framework to a more complex analysis of the interaction between visual, linguistic, and auditory information, while preserving its competence-oriented perspective. The study considers two complementary representations of audio: raw signal and automatic transcriptions provided as textual input. By evaluating these representations both independently and in combination with video information, the analysis investigates whether audio contributes meaningfully to task performance and whether one representation has a stronger influence on model predictions. This leads to two research questions:

RQ1: *To what extent does raw audio affect video-based Visual Statement Verification when it is jointly processed with visual and textual information?*

RQ2: *How effectively do multimodal models integrate visual, textual, and auditory information when performing video-based reasoning and understanding tasks?*

The experiments show that raw audio has a limited and strongly context-dependent effect. It is most useful when visual information is unavailable, whereas, when added to an already informative visual input, it may provide little benefit or introduce noise. The results also show that transcriptions can produce greater interference than raw audio, suggesting that multimodal architectures do not necessarily achieve more effective cross-modal understanding simply because additional modalities are available. These findings must nevertheless be interpreted in relation to the nature of the dataset. The one hundred original MAIA videos were selected and annotated according to their visual content, and the questions were formulated without considering the audio track. Consequently, the benchmark does not systematically require auditory information to solve its items. My thesis project begins from this limitation. The following chapters investigate whether models can integrate multiple input modalities more coherently when the dataset is specifically designed so that auditory information is at least partially relevant. The central hypothesis is that, under these conditions, models may use complementary signals across modalities to reconstruct missing information and develop a more genuinely multimodal understanding of video content, producing answers grounded in the available evidence rather than in linguistic plausibility or chance.

1. Literature review

The rapid evolution of Visual Language Models (VLMs) has progressively expanded the capacity of neural systems to jointly process perceptual and linguistic information. Underlying this line of research is the hypothesis that linguistic meaning cannot be constructed exclusively through distributional regularities learned from text corpora, but rather needs to be grounded in perceptual experience as well[2]. In this context, the Visual Question Answering (VQA) task represents one of the first systematic paradigms to verify whether a model is capable of answering questions referring to an image's content [1]. Subsequent benchmarks, such as **Hudson and Manning's (2019)**, have expanded upon this setting by incorporating compositional questions related to entities, properties, and visual relations [7]. The elevated accuracy achieved does not imply a sufficient foundation to assess multimodal comprehension. Several studies demonstrate that models can exploit statistical correlations present in the answers, in the questions, or in dataset distributions, yielding the correct prediction even without *understanding* the visual content. Therefore, evaluation has progressively shifted from measuring merely task performance to verifying grounding, defined as the ability of a model to associate a prediction with relevant perceptual evidence. The most prevalent grounding assessment method consists of employing minimally contrastive pairs. In this specific setting, the correct decision is paired with a foil, obtained by modifying one semantically relevant element. *FOIL-it!* adopts this formulation to evaluate whether the model recognizes localized discrepancies between an image and its linguistic description [13]. Conversely, *Winoground* proposes pairs of images and descriptions containing the same lexical elements, but organized according to different compositional relations, demonstrating that the recognition of individual entities does not necessarily imply the comprehension of their relational structure [17]. Shifting from image to video inputs introduces additional challenges. Video comprehension requires integrating information diluted across time, recognizing actions and state changes, ordering events, and inferring temporal and causal relationships. *Action Genome Question Answering (AGQA)* addresses this problem by utilizing compositional questions referring to objects, actions, and spatio-temporal relations [6], whereas the *Multi-modal Video Understanding Benchmark (MVBench)* organizes its evaluation around a broader set of competencies, including action perception, movement, temporal order, and dynamic reasoning [10]. However, even with video benchmarks, models can cut corners by exploiting unimodal shortcuts or relying solely on the most informative frame. **Kesen et al. (2024)**[8] challenge this limitation by proposing *Video Language Model Assessment (VILMA)*, a zero-shot benchmark based on video-text contrastive pairs. For each complex competency, the benchmark introduces a proficiency test to verify the preliminary perceptual capacity needed to solve the task. The results demonstrate that, once those preliminary conditions are established, a portion of the apparently correct answers can be attributed to accidental or spurious strategies. In particular, the VLMs examined do not exhibit any systematic advantage over statistical image-based models in tasks requiring strict temporal grounding. With MAIA, a new natively italian benchmark was introduced, aiming to provide a fine-grained evaluation of video-linguistic reasoning [16]. This resource encompasses twelve reasoning categories related to linguistic phenomena: causal, counterfactual, implicit, spatial, temporal, sentimental, planning, uncertainty, and out-of-scope. The innovation of MAIA consists in aligning two tasks built upon the same videos and questions: *Visual Statement Verification (VSV)*, in which the model must select the correct statement from a minimally contrastive pair, and *Open-Ended Visual Question Answering (OEVQA)*, in which the model is required to freely generate the correct answer. The setting of the MAIA benchmark allows us to assess not only accuracy, but also coherence between discriminative comprehension and linguistic generation. A model might answer a multiple-choice question correctly, yet be unable to reproduce the same information in an open-ended task, furthermore, it may exhibit unstable behavior across equivalent linguistic formulations. The *Aggregate Accuracy* metric proposed by MAIA rewards only instances in which the model answers correctly on all true-false pairs associated with the same question and, simultaneously, generates the correct answer for the related open-ended question task. The observed reduction in Aggregate Accuracy compared to traditionally measured single-case accuracy highlights the fragility of apparently high performance, as well as the necessity to jointly evaluate robustness, consistency, and grounding. Although VLMs are traditionally built upon textual-visual interactions, real videos are intrinsically audiovisual. The meaning of video content can depend on a broad array of multimodal factors, such as music, prosody, background noise, and other acoustic signals that cannot be reconstructed solely from visual frames. For this reason, an increasingly large portion of research extends multimodal language model architectures to enable cross-modal processing among text, images, videos, and audio. A new proposal emerged with *ImageBind*, which introduced the concept of a representation space shared among images, text, audio, depth, thermal information, and inertial movement [5]. Other multimodal language models (MLMs) employ specific encoders for each modality along with projection or fusion modules to linguistically align these representations. *Video-SALMONN:Speech-Enhanced Audio-Visual Large Language Models* introduces a multi-resolution Q-former[1] to align audio signals with synchronized video [14], whereas *VideoLLaMA2* combines spatio-temporal modeling and acoustic processing within a generative multimodal architecture [3]. Contemporary omnimodal architectures process all different modalities within a single unified framework. *Qwen2.5-Omni* uses a thinker-talker structure that accepts text, image, video, and audio inputs to generate textual or audio responses [11]. The next step in the Qwen-Omni evolution is *Qwen3.5-Omni*, which integrates visual and audio encoders into a common backbone and adds explicit temporal references to improve the alignment between audio and video [12]. This architectural evolution demonstrates that it is technically possible to simultaneously process heterogeneous signals. However, the availability of a modality does not imply that it is used in a functional and complementary way. There is an urgent need to articulate this distinction between **multimodal availability** and **multimodal integration**: the former merely implies that the system can accept multiple input sources, whereas the latter implies that the model selects, coordinates, and combines relevant information from each modality. This means that a system that is multimodal from an architectural point of view might not behave as such, predominantly relying on unimodal information during prediction. This inconsistency clearly emerges in *Audio-Visual Question Answering*

[1]A multi-resolution Q-former, like a Multi-Resolution Causal Q-Former, is a sophisticated component in multimodal architectures. It manages continuous audio, video, or image data at various temporal or spatial scales using distinct query banks to balance detailed signals with broad context.

benchmarks, such as MUSIC-AVQA, which assesses spatio-temporal relations in musical performance videos [9], and *AVQA* [9], which extends the analysis to objects, entities, and everyday activities. Subsequent studies highlight a strong bias toward visual information, demonstrating that the majority of the questions in the task can be answered solely from the video stream without utilizing the audio input channel. Under this assumption, the similarity between audio-only performance and video-only performance does not prove genuine integration, instead, it evidences a form of unimodal collapse that dominates the decision-making process.

2. Approfondimento o ipotesi

Analisi dettagliata del fenomeno specifico ed esposizione delle ipotesi di ricerca.

3. Materiali e metodi

4. Results

5. Future Directions

Conclusioni

Conclusioni...

Bibliography

[1] Stanislaw Antol et al. "VQA: Visual Question Answering". In: *Proceedings of the IEEE International Conference on Computer Vision*. 2015, pp. 2425–2433.

[2] Marco Baroni. "Grounding Distributional Semantics in the Visual World". In: *Language and Linguistics Compass* 10 (2015). DOI: 10.1111/lnc3.12170.

[3] Zesen Cheng et al. "VideoLLaMA 2: Advancing Spatial-Temporal Modeling and Audio Understanding in Video-LLMs". In: *arXiv preprint arXiv:2406.07476* (2024). arXiv: 2406.07476. URL: https://arxiv.org/abs/2406.07476.

[4] Michela Deodati et al. "Lost in Transcription: Investigating the Role of Audio Information for Video-Based Visual Statement Verification". In: *Proceedings of the Twelfth Italian Conference on Computational Linguistics (CLiC-it 2026)*. Forthcoming. 2026.

[5] Rohit Girdhar et al. "ImageBind: One Embedding Space to Bind Them All". In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2023, pp. 15180–15190.

[6] Madeleine Grunde-McLaughlin, Ranjay Krishna, and Maneesh Agrawala. "AGQA: A Benchmark for Compositional Spatio-Temporal Reasoning". In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2021, pp. 11287–11297.

[7] Drew A. Hudson and Christopher D. Manning. "GQA: A New Dataset for Real-World Visual Reasoning and Compositional Question Answering". In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2019, pp. 6700–6709.

[8] Ilker Kesen et al. "VILMA: A Zero-Shot Benchmark for Linguistic and Temporal Grounding in Video-Language Models". In: *International Conference on Learning Representations*. 2024. arXiv: 2311.07022. URL: https://arxiv.org/abs/2311.07022.

[9] Guangyao Li et al. “Learning to Answer Questions in Dynamic Audio-Visual Scenarios”. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2022, pp. 19108–19118.

[10] K. Li et al. “MVBench: A Comprehensive Multi-Modal Video Understanding Benchmark”. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2024. DOI: 10.48550/arXiv.2311.17005. arXiv: 2311.17005.

[11] Qwen Team. “Qwen2.5-Omni Technical Report”. In: *arXiv preprint arXiv:2503.20215* (2025). arXiv: 2503.20215. URL: https://arxiv.org/abs/2503.20215.

[12] Qwen Team. “Qwen3.5-Omni Technical Report”. In: *arXiv preprint arXiv:2604.15804* (2026). arXiv: 2604.15804. URL: https://arxiv.org/abs/2604.15804.

[13] Ravi Shekhar et al. “FOIL it! Find One Mismatch between Image and Language Caption”. In: *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 2017, pp. 255–265. DOI: 10.18653/v1/P17-1024. URL: https://aclanthology.org/P17-1024/.

[14] Guangzhi Sun et al. “Video-SALMONN: Speech-Enhanced Audio-Visual Large Language Models”. In: *arXiv preprint arXiv:2406.15704* (2024). arXiv: 2406.15704. URL: https://arxiv.org/abs/2406.15704.

[15] Davide Testa et al. “All-in-One: Understanding and Generation in Multimodal Reasoning with the MAIA Benchmark”. In: *Findings of the Association for Computational Linguistics: EMNLP 2025*. 2025, pp. 20030–20050. DOI: 10.18653/v1/2025.findings-emnlp.1091. URL: https://aclanthology.org/2025.findings-emnlp.1091/.

[16] Davide Testa et al. “MAIA: A Benchmark for Multimodal AI Assessment”. In: *Proceedings of the Eleventh Italian Conference on Computational Linguistics (CLiC-it 2025)*. 2025, pp. 1121–1134. URL: https://aclanthology.org/2025.clicit-1.106/.

[17] Tristan Thrush et al. “Winoground: Probing Vision and Language Models for Visio-Linguistic Compositionality”. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2022, pp. 5238–5248.